MIL-B-52097A(ME)
7 January 1975
SUPERSEDING
MIL-B-52097(CE)
8 March 1960

MILITARY SPECIFICATION

BARS, CARRYING: MULTIPLE AND SINGLE, M4 BALK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers multiple and single carrying bars for the M4 balk.

1.2 Classification. Carrying bars shall be of the following types, as specified (see 6.2):

Type I - Multiple balk carrying bar.
Type II - Single balk carrying bar.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-T-704	- Treatment and Painting of Materiel.

FSC 5420

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspecition by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

13220E6843	- Bar, Carrying: Multiple Balk.
13220E6844	- Bar, Carrying: Single Balk.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC. (AWS)

A5.5 - Low-Alloy Steel Covered Arc-Welding Electrodes.
D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington DC 20036.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Description. Carrying bars shall be as shown on the following drawings and as specified herein:

Type I - 13220E6843
Type II - 13220E6844

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the carrying bar. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one carrying bar for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce carrying bars that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as shown on the applicable drawings and as specified herein.

3.3.1 Welding electrodes. Welding electrodes shall conform to AWS A5.5 Classification E7015 or E7016, and size as required.

3.4 Identification marking. The equipment shall be identified in accordance with MIL-STD-130.

3.5 Treatment and painting. The bars shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B.

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3.6 Workmanship.

3.6.1 Metal fabrication. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.6.2 Welders and welding.

3.6.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME, Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Suppliers who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government the supplier's standard welder qualification may be substituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

3.6.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Soceity publication "D1.1-Structural Welding Code". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld matal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.6.2.3 Welding jigs. Components shall be assembled in steel jigs or frames and welded while held in position. The jigs or frames shall be of a design that will limit the distortion to an amount that can be removed without changing the physical characteristics of the metal of the parts being welded.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction carrying bar shall be examined for defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The carrying bar(s), as applicable, shall be examined as specified herein for the following defects:

Major

101. Dimensions not in accordance with drawings.
102. Material not as specified.
103. Incorrect assembly.
104. Edges not round.
105. Weld defects such as base-metal cracks adjacent to or behind welds, crater cracks or cracks in weld metal, spatter, undercutting of base metal, welding distortions, or hooks not in alinement.

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Minor

201. Identification marking incomplete or illegible.
202. Treatment and painting not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Type I bars not bundled and secured as specified for Level A or B.
107. Strapping not as specified for Level A or B.
108. Boxes not as specified for Level A or B.
109. Blocking and bracing not adequate to prevent movement or damage for Level A or B.
110. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I bars. The bars in quantities of four shall be bundled. Each bundle shall be secured with not less than three equally spaced straps conforming to QQ-S-781, Type I or IV, regular duty, Finish B, Size 1/2 inch wide and 0.015 thick; seals as applicable.

5.1.1.2 Type II bars. The bars shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, or PPP-B-621, Class 2, style optional, in quantities not to exceed the weight limitation of the box. The bars shall be blocked and braced within the box to prevent face movement or damage. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

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5.1.2 Level B. Packing shall be as specified in 5.1.1 for Level A except boxes for Type II bars shall be Domestic Type or Class as applicable.

5.1.3 Level C. The bars shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings. Containers shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Bundles and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type I carrying bar is issued as part of a sling to lift a group of M4 balk simultaneously in the assembly of fixed and floating bridges. The Type II carrying bar is used to facilitate manual handling of M4 balk in the assembly of fixed and floating bridges.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of bar required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production carrying bars from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish carrying bars conforming to this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-B-52097A(ME) Bars, Carrying: Multiple and Single, M4 Balk

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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